

Design Agency in Corporate Strategy

December, 2024

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Research Question:

1. How can Design Agency enable agentic intelligence within corporations, fostering strategic resilience and transformative impact in response to the speed and scale of global challenges?

Current Research Hypothesis:

Integrating design agency as a multi-intelligence model, combining human and machine agency, fosters a new form of collaboration within corporate settings, enhancing decision-making, driving adaptability to complex challenges, and leading a competitive advantage.

Research Summary

This research explores the evolution of design competencies into Design Agency within corporate ecosystems, emphasizing their transformation through dynamic interactions with intelligent systems. Anchored in theoretical and participatory action research methodology, it examines how agentic design capabilities emerge through coupling and decoupling processes across system, strategy, and product levels. By integrating human agency, algorithmic thinking, and large language models (LLMs), the study reframes design competencies as a multi-intelligence capability, opening new possibilities for the design practice within complex corporate environments while addressing complex challenges with enhanced scalability and precision.

Knowledge gap

In recent decades, design practices, particularly human-centered design, have primarily concentrated on the context of consumption, often overlooking how design adapts to the broader institutional context in which it operates. As John Heskett (1998) argues, design is a crucial economic force, acting both as an agent of change and a catalyst for value creation within businesses. Heskett positions design as a broker and translator, bridging the often-disconnected domains of production and consumption. By aligning the operational demands of production with the evolving needs of consumers, design enables innovations that are simultaneously market-responsive and operationally viable.

However, this consumption-driven focus has limited organizations' understanding of design's broader strategic potential within other institutional contexts, such as its role in shaping and being shaped by production processes, corporate governance, strategic frameworks, and investment priorities. This narrow application has created a substantial knowledge gap, particularly in understanding design capabilities as dynamic agents augmented by a technological advanced context. Today's rapidly evolving business environment, marked by increasingly complex, interconnected challenges, demands a new operational model for design, one that transcends the traditional bounds of human cognition and operational constraints.

Rather than confining design to the company level, this research reframes design as a core driver of growth, operational adaptability, and competitive advantage across a conglomerate's varied subsidiaries. By exploring how design operates within a post-industrial production context and expands as a multi-intelligent capability within the vast organizational ecosystems, this study proposes a transformative model for design practice. This approach not only fills a critical gap but also repositions design as an indispensable component in navigating the future challenges of conglomerates.

Research Process

The overarching research methodology is shaped by two foundational blocks that guide its structure and intent. The first draws on the rich tradition of collaborative design research, evolving from Action Research in the 1940s, where researchers engaged directly with real-

world environments to drive change, to Participatory Research in the 1970s, which brought an emphasis on integrating the perspectives and experiences of participants into the research process. Building on these foundations, contemporary approaches such as Speculative and Practice-Based Research have emerged, allowing design research to push boundaries through experiential inquiry and hypothesis-driven experimentation. Together, these methods establish a foundation that is both collaborative and exploratory, enabling this research to engage deeply with the real-world dynamics within corporate settings.

The second centers on a rigorous, iterative process of deconstructing and reconstructing the thesis theoretical framework. This approach allows for a precise examination of each component, enabling the organic emergence of potentially paradigm-shifting concepts. By integrating established design frameworks with complementary theories, this research builds a dialog that functions as an independent theoretical space. This space not only validates emerging insights but also initiates the groundwork for the hypothetical Design Agency Model, a framework that will be explored further later in this report.

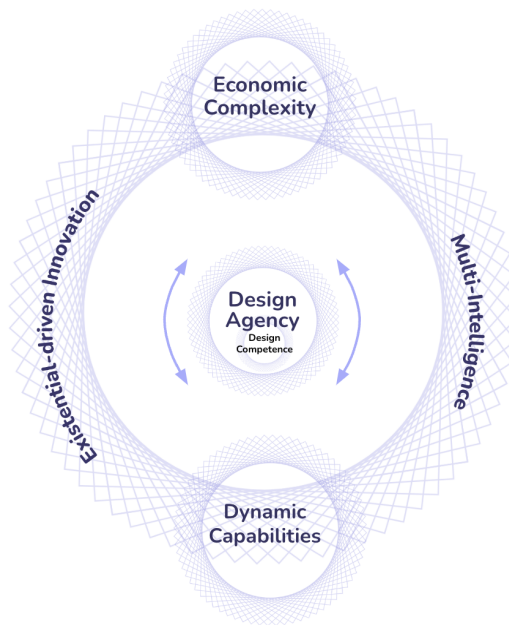


Figure 1: Theoretical Framework

The process involves conducting interviews, a semester-long program at ID, case studies, and multiple episodes with semi-controlled experiments:

- A total of eight interviews were conducted with various leaders and roles within Credicorp to gain insights into how design competencies interact across the organization and to identify the emergence of Design Agency.
 - These interviews served as a guiding umbrella, providing key inputs that informed the development of the studies and tools presented in this research.
 - They were instrumental in shaping terminology, building a shared understanding among multiple stakeholders, and exploring the dynamic interplay between design and other business functions. They highlighted specific intersections where design begins to exert agency, influencing corporate processes and decision-making.
 - Details of the raw data, data analysis and data samples derived from these interviews can be found in the associated folder. All studies were developed within the overarching context of these conversations, ensuring alignment with the realities and complexities of the corporate environment.
 - Data sample:



Figure 2: Interviews sensemaking process

- **ID Multi-Intelligence Design course:** A semester-long program conducted with master's students at the Institute of Design, explores the integration of Language Models (LLMs) and algorithmic thinking within design practices. Through a series of experiments using diverse tools and frameworks, students engage in reimagining

existing workflows and design frameworks, leveraging the transformative potential of AI.

- The course was structured around two corporate briefs provided by Credicorp as part of a collaborative partnership. Financial Literacy (Annex 1) and Sustainability and Inclusion in the Health Industry (Annex 2).
- Students were tasked with conducting five experiments that explore three core approaches and concepts: strategic choice-making; algorithmic futuring, and design agency. These experiments investigated how design functions as a multi-intelligent competency within complex organizational settings, examining existing workflows and design frameworks while harnessing AI's transformative potential.
- **Case Studies:** This research is structured around two case studies, each situated within a corporate setting to examine the role of design agency as a strategic, multi-intelligent capability within large organizations:
 - **Case study 1:** This case study explores the transformation of UX Writing (UXW) at Yape, a leading Peruvian payment platform, from a human-centric practice to a hybrid, multi-intelligent competency. By integrating human expertise with AI-driven agents, UXW emerges as a dynamic capability that enhances efficiency, consistency, and adaptability within Yape's expansive digital ecosystem.
 - **Case study 2:** This study explores the process of granting agency to Research as a design competency within BCP, Peru's largest bank under Credicorp. The case explores how this intelligent research agent, trained in frameworks and research capabilities, interacts with other entities, including knowledge experts operating with similar frameworks. It investigates how these interactions enable autonomous operations, foster collaboration, and maintain strategic alignment with corporate innovation goals, creating an agile and adaptive design ecosystem.
- **Episodes:** In the context of the case studies, each episode serves as a structured, experiential unit of inquiry that enables in-depth exploration of design competencies in action within real-world scenarios. It functions as a controlled setting where design agency is observed and tested through interactions with various entities, agents, and contexts.

Building blocks of discoveries: Studies

As part of this exploration, a series of studies emerged. Our studies represent a series of building blocks of discoveries interconnected yet distinct, each targeting a foundational aspect of the broader inquiry to uncover patterns and principles

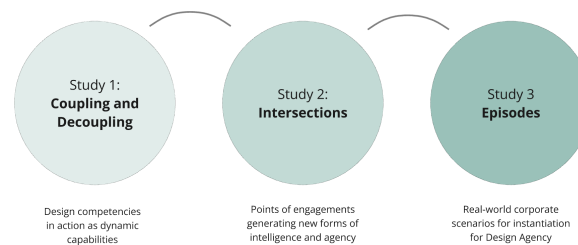


Figure 3: Studies

1. **Study 1: Coupling and decoupling of design competencies.** This study emerged as a natural extension of the action research process, focusing on the dynamism of design competencies. By exploring their interactions with various entities and contexts, the study uncovers the mechanisms of coupling and decoupling processes through which design competencies integrate into systems to tackle challenges (coupling) and disengage to promote adaptability and innovation (decoupling). While this research has primarily observed these dynamics within the context of production, it posits that similar mechanisms operate within the contexts of use and investment, offering a comprehensive framework to understand how design competencies evolve into adaptive, agentic forces within corporate environments.c

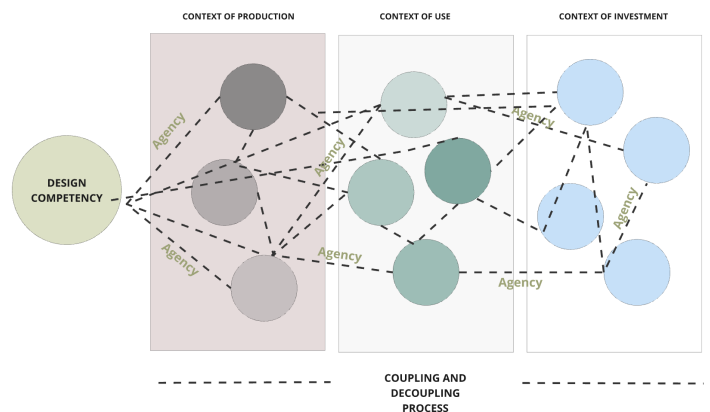


Figure 4: Coupling and decoupling agentic design process

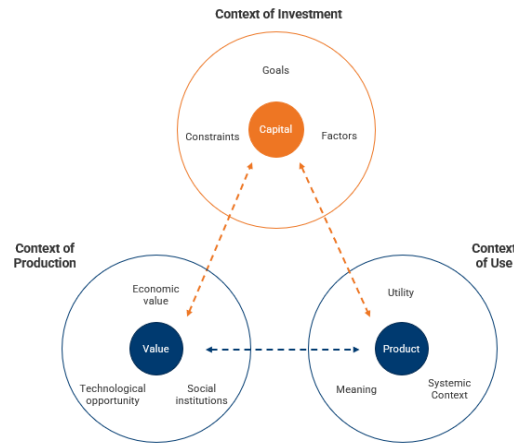


Figure 5: Teixeira 2002; Heskett, 1998, Ostapchuk, 2024

2. **Study 2: Intersections.** The study of "intersections" emerges from understanding how design competencies in action navigate within corporate contexts, engaging with diverse entities such as algorithmic thinking, corporate frameworks, and organizational roles. These intersections are not merely points of contact but dynamic spaces where competencies are activated, evolve, and gain agency through action. By observing how design competencies operate across organizational levels such as System, Strategy, and Product, this study uncovers how these intersections generate new forms of intelligence and value. Design agency not only arises from these interactions but also actively creates new intersections, reshaping how competencies are utilized and how value is produced, ultimately redefining the strategic role of design within corporate structures.

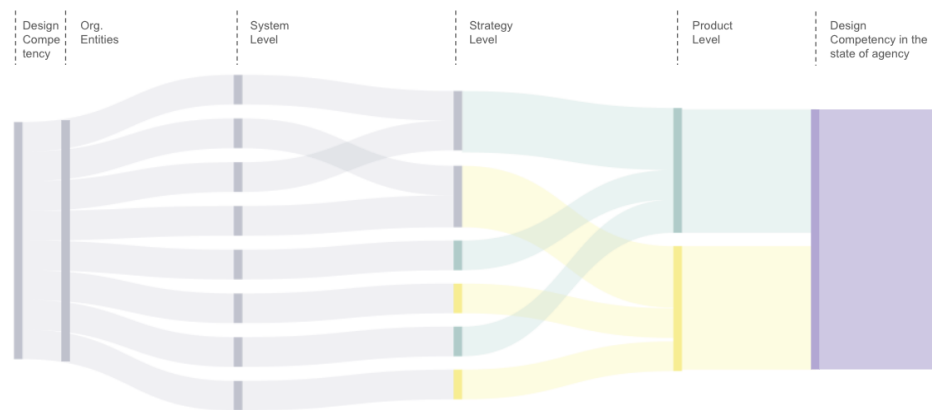


Figure 6: Intersections Mapping

3. **Study 3: Episodes.** This study arises from the need to instantiate the studies of coupling, decoupling, and intersections into practical, real-life contexts. An episode functions as a structured, experiential unit of inquiry designed to demonstrate these dynamics within corporate environments. By situating design competencies in action, episodes provide a controlled setting to explore how they interact with various entities, agents, and contexts, evolving into agentic states. Through observation, testing, and refinement, episodes reveal the transformative potential of design agency in addressing complex organizational challenges.

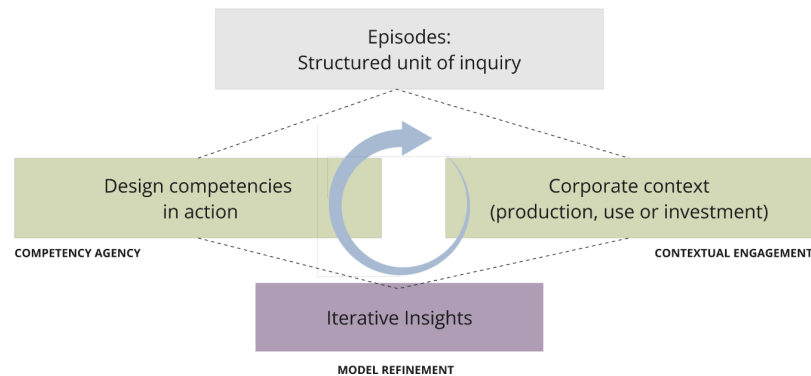


Figure 7: Episodes iterative framework

Participatory Action Research: Data Analysis and Emerging Insights

The unit of analysis in this research is *design competencies in action*, focusing on how these competencies create intersections with multiple entities within a coupling and decoupling process. Observing specific design competencies across episodes within the stages of a solution development flow offers a structured lens to examine their evolution, particularly as they engage with five distinct intelligence configurations: Human (singular), AI (singular), Human x Human (Collective), Human x AI (Hybrid), and AI x AI (Hyper). These configurations enable the evaluation of diverse approaches to problem-solving, adaptability, and decision-making, shedding light on how design agency emerges and achieves impact through these interactions.

Intelligence Model	Solution Development Flow			
	Analysis	Synthesis	Prototype	Strategize / Decision Making
1 Human (singular)	Episode			
2 AI (singular)	Episode			
3 Human x Human (Collective)	Episode			
4 Human x AI (Hybrid)	Episode			
5 AI x AI (Hyper)	Episode			

Figure 8: Intelligence models across episodes

During the participatory research process, we defined the components, variables, and metrics that would establish the structural foundation of the episodes. This involved articulating the core elements that constitute an episode, pinpointing the factors influencing the transformation of design competencies into agency, and establishing measurable indicators to assess outcomes. These foundational elements ensured that each episode could systematically explore the coupling and decoupling processes, the generation of intersections, and the impact of multi-intelligent configurations on the evolution of design competencies.

Component	Description	Variables to Manipulate	Evaluation Metrics	Questions
Objective	Defines the specific goal of the episode within the larger framework of testing design agency in a multi-intelligent model.	Type of design task (e.g., UX Writing, Prototyping)	Achievement of task goal Quality of output	What specific outcome is the episode trying to achieve? Does the task require human insight, AI support, or both?
Intelligence Configuration	Determines which intelligence (Human, AI, Hybrid) are involved and their roles.	Human-only, AI-only, Human x Human, Human x AI and AI x AI (Hybrid)	Effectiveness of interactions	How does the chosen intelligent configuration impact task completion? What roles do humans and AI agents play, and how do they influence each other?
Task Structure	The structured workflow or steps each agent follows to complete the design task.	Sequence of steps Automation level	Process efficiency Adaptability of workflow	What steps are involved in the workflow? How automated is each step, and does it allow flexibility for iterative improvement?

Input	Initial data, guidelines, or instructions provided to agents to begin the task.	Specificity of input Relevance of context	Completeness of input usage Accuracy of execution	How does the specificity of input affect task performance? Is the information provided enough for the agents to understand and execute the task effectively?
Process	The internal steps, actions, and interactions between agents as they work on the task.	Degree of autonomy Type of collaboration	Task progression Quality of collaboration	How do agents interact throughout the process? What challenges or strengths arise from autonomous vs. collaborative efforts?
Output	The final deliverable or result of the episode's design task, reflecting the quality and accuracy of agent-driven work.	Format Relevance to original task	Quality of deliverable User satisfaction	Does the output meet the intended objectives? How does the output compare to expectations for quality and usability?

Drawing on a combination of expert interviews, in-depth observations, and my active role as Corporate Head of Design and Experience at Credicorp, a leading financial holding in LATAM, this study leverages a dual perspective as both practitioner and researcher to deeply examine how design competencies are identified, operationalized, and evolved within an organizational context.t. The findings revealed that design competencies operate across three distinct organizational levels: System, Strategy, and Product, with specific competencies identified at each level. At the System level, competencies address overarching frameworks and policies, aligning design with corporate governance and operational coherence. At the Strategy level, they guide decision-making processes, shaping value propositions and aligning design efforts with business objectives. Finally, at the Product level, competencies are operationalized through tangible outputs like UX writing, prototyping, and content design. This multi-level perspective lays the groundwork for understanding how design competencies transition into design agency through dynamic interactions and intersections within these levels.

Organizational Levels	Design Competency	Description
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System Level	System Thinking	Enables holistic understanding of interdependencies within corporate ecosystems and between internal and external stakeholders.
	Organizational Foresight	Identifies emerging trends and potential disruptions, supporting proactive adjustments to system-wide processes and policies.
	Policy Design	Develops frameworks that guide design initiatives within regulatory and organizational standards.
	Cultural Change Facilitation	Drives shifts in organizational culture to prioritize design as a strategic asset.
	Complex Problem-Solving	Addresses multifaceted challenges that span various functions, using design to align organizational goals and capabilities.
Strategy Level	Strategic Choice-Making	Supports executive decisions by providing structured design inputs for balancing innovation with risk management.
	Brand Positioning and Identity	Crafts a cohesive brand identity that aligns with long-term organizational goals and resonates with target markets.
	Value Proposition Design	Develops value propositions that differentiate the corporation and meet evolving market demands.
	Innovation Pipeline Development	Establishes a systematic approach for continuous innovation, integrating design at all stages.
	Stakeholder Alignment	Ensures that design efforts align with the interests of key stakeholders, fostering support and buy-in.
Product Level	User-Centered Design	Prioritizes user needs and preferences to create products that enhance user experience and satisfaction.
	Prototyping and Testing	Rapidly iterates on product ideas to refine usability, functionality, and appeal based on real-world testing.
	UX Writing and Content Design	Crafts effective messaging that guides and enhances user interactions with products.

	Aesthetic and Functional Design	Balances aesthetic appeal with practical functionality to meet user expectations and brand standards.
	Market Feedback Integration	Incorporates feedback from users and market trends to continuously improve and adapt products.

We observed significant tensions as traditional design competencies evolve into design agency. This transformation challenges established practices, as competencies once confined to specific tasks now expand to address broader, system-wide objectives, requiring a more adaptive and computationally powered approach. The transition to a multi-intelligent, agentic model unveils both opportunities and complexities.. The following table highlights these shifts and the tensions they generate.

Tension	When Using Design Competencies	When Shifting to Design Agency
Focus and Scope	Narrow focus on task-specific, project-based goals within defined parameters.	Expands to a broader, strategic role, aiming to influence long-term organizational outcomes and systemic transformation.
Intelligence Configuration	Relies on human expertise, intuition, and manual processes limited by cognitive constraints.	Incorporates multi-intelligent inputs (human insight + algorithmic thinking and computational power), enabling richer data-driven decision-making and pattern recognition.
Adaptability	Constrained within established frameworks and methods, limiting flexibility and adaptability to rapid change.	Adopts fluid frameworks, enabling dynamic coupling and decoupling with various systems to adapt continuously to new organizational needs.
Role in Decision-Making	Primarily consultative or advisory, providing structured inputs without authority in strategic decisions.	Becomes a proactive decision-making agent with a direct influence on strategic priorities and organizational direction.
Speed and Scalability	Dependent on human capacity, resulting in limited speed, scalability, and the need for manual iteration.	Enhanced by AI-driven processes, allowing rapid iterations and scalability across multiple organizational contexts.

Knowledge Creation	Produces siloed project-based knowledge through human-driven research and insights that may become outdated over time.	Generates dynamic, evolving knowledge by combining human insights with continuous AI data integration and real-time feedback loops.
Resource Dependency	Dependent on specialized design teams with dedicated resources, often confined to specific project budgets and timelines.	Leverages distributed resources and cross-departmental knowledge, accessing expertise across the organization as needed.
User-Centered vs. System-Centered	Primarily focused on individual user needs and product usability, often with a narrow perspective on the system	Balances user needs with broader corporate objectives and system-wide impacts, integrating user insights with strategic and operational goals.

Understanding Core Tensions

- **Focus and Scope:** Traditional competencies are limited to task-based outcomes, while agency extends design's role into long-term, strategic influence, potentially creating tension between immediate project needs and overarching organizational goals.
- **Intelligence Configuration:** Design competencies rely solely on human insights, whereas design agency integrates multi-intelligent inputs, enabling richer, more accurate insights but potentially creating conflicts with designers who may feel replaced or diminished.
- **Adaptability:** Competencies follow structured methods, which limits adaptability. Shifting to agency introduces fluid frameworks, allowing competencies to evolve as organizational needs change, which may cause friction with those favoring established processes.
- **Role in Decision-Making:** Traditional design competencies are often advisory, while agency takes a proactive, decision-influencing role. This shift may cause resistance from leadership, who may hesitate to cede decision-making power to design.
- **Speed and Scalability:** Competency-based design operates at the pace of human work, while agency uses computational power and algorithmic thinking to increase

speed and scalability. This creates tension between the need for deliberate human-centered approaches and AI-enabled rapid iteration.

- **Knowledge Creation:** Competency-based design produces insights that may quickly become outdated or not shared and evolved. Design agency uses real-time data and artificial intelligence for dynamic knowledge creation, challenging designers to stay up-to-date and adapt quickly.
- **Resource Dependency:** Competencies rely on dedicated resources for each project, while agency uses a distributed, shared-resource model. This can stretch traditional design resources and cause conflicts over ownership and autonomy.
- **User-Centered vs. System-Centered:** Competencies focus on individual user needs, while agency balances this with system-wide objectives, creating tension between user experience goals and larger strategic imperatives within the corporation.

Case Studies and Episodes

Case Study 1: Leveraging Agency-Driven UX Writing to Transform User Experience at Yape

Episode: Human x AI Hybrid Intelligence

This episode delves into the effectiveness, limitations, and transformative potential of UX Writing as it transitions into a state of agency. It explores how integrating intelligent systems enhances or constrains the design process, particularly in addressing user frustration while maintaining brand values and strategic alignment.

Context

Yape, a payment platform aiming to evolve into a super app, enables users to perform diverse transactions, including shopping. Communication within this ecosystem is critical, as it serves a massive and diverse user base. This episode explores how a UX Writing designer interacts with an AI agent trained as a knowledge expert to create empathetic and effective communication content.

Technical aspects:

- Participant: A UX Writing design expert with 5 years of experience in crafting digital content for user interfaces
- Tools and software: Microsoft Co-Pilot Suite (free version). Utilization of GPT-based AI for content generation and refinement.
- Data integration: AI agent trained on UX Writing career path frameworks, brand style guides, and a prompt system tailored to align its output with Yape's strategic goals and brand identity.
- Interaction workflow: The UX Writer collaborates with the AI agent through a co-creative interface, iteratively refining content suggestions.

Objective of the episode

- Analyze the process: Observe how UX writing competency in action evolves into agency through the coupling and decoupling process when interacting with different entities, humans and no humans.
- Identify intersections: Map the points where the design competency generates meaningful intersections with entities.

Activities

1. Introduction & onboarding: Explain the context and goal of the session.
2. Baseline task: Assign a UXW task (e.g., creating an "unhappy" microcopy) that participant will complete collaboratively with the AI agent.
3. Intersections map: Mapping how the UX Writing competency interacts with various entities (e.g., brand guidelines, product designers) and observing how these interactions generate intersections that transform sub-competencies into agentic states.
4. Filing the reflection log: Participants fill a form to capture their qualitative insights after completing the episode, focusing on their experiences, challenges, successes, and areas of improvement when engaging with the Hybrid Intelligence model for UX Writing tasks.
 - a. Task overview: Participants will share their approach to collaboration with the AI agent.

- b. Tensions observed: Participants note points of friction or challenges they faced
- c. Highlights identified: Participants document moments of success or synergy
- d. Intersections and transformations: Participants reflect on how the task revealed intersections between human expertise, AI capabilities, and organizational frameworks.

Data collected

- UXW Career Path
- Prompt System: Custom-built scenarios and interaction templates that enable the AI to generate contextually relevant content, adhering to strategic priorities.
- Baseline task
- Intersection mapping
- Final piece and impact
- Reflection log

Data samples

- **Baseline Task: Unsuccessful Payment for a Service**

"Dani, we need a copy for an unsuccessful payment scenario for users trying to pay for a service. When they reach the final stage, the service provider is unresponsive for some reason, so the user cannot complete the payment. The idea is to provide them with an alternative option."

Requerimiento: Unhappy al pagar un servicio

"Dani, necesitamos un copy para un unhappy para aquellos usuarios que están intentando pagar un servicio y ya cuando están en la etapa final, el proveedor del servicio no seta respondiendo por alguna razón, entonces el yapero no puede hacer el pago. La idea es darla alguna opción"

Figure 9: The task presented to the UX writer participant

To accomplish this task, the participant leverages a hybrid intelligence model (Human x AI), using Microsoft Co-Pilot as the AI agent trained in Yape's UXW guidelines, brand frameworks, and business logic.

- **Intersection Mapping Tool:** The Intersection Mapping Tool emerges from the study of intersections discussed earlier. This tool was developed as a structured framework to analyze and visualize the dynamic engagements between a design competency and other organizational entities. The tool allows us to highlight how competencies transition into agentic states.

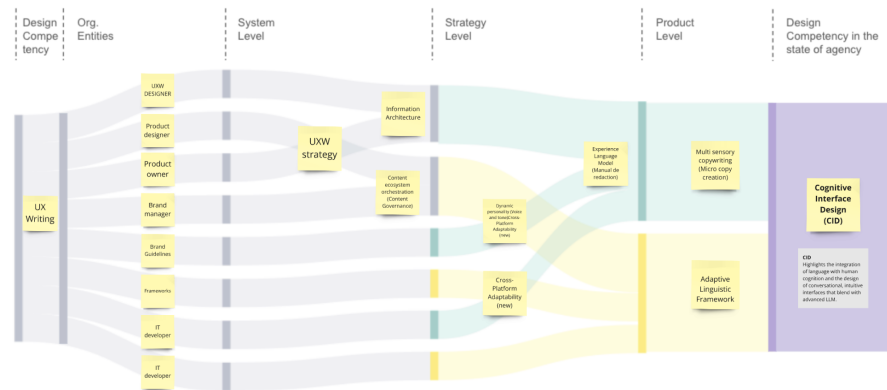


Figure 10: Intersections Map

- **Impact in time reduction:** By applying the coupling and decoupling process, we observe significant reductions in the time required to execute design-related tasks. This dynamic framework allows for the rapid integration and reconfiguration of design competencies, ensuring that the right resources are deployed at the right moments. As a result, iterative cycles, such as those in UX Writing, are streamlined, reducing bottlenecks and accelerating the delivery of scalable, user-centered solutions aligned with strategic goals.



Figure 11: Impact in productivity

Key takeaways

- **Coupling and Decoupling as a Mechanism to Operate Design Agency:** We observed that this mechanism underpins the operationalization of Design Agency, allowing it to adapt to shifting contexts and needs.
- **Agentic Evolution of Design Competency:** The UX Writing episode demonstrated how a traditionally design competency evolved into an adaptive, agentic capability. Through iterative interactions with organizational entities, UX Writing in action was able to integrate effectively across system, strategy, and product levels, driving a result that aligned with the company's strategic objectives.
- **Emergence of a New Form of Design Competency:** This episode reveals the emergence of a hybrid, multi-intelligent design competency, capable of merging human creativity with algorithmic intelligence. This represents a significant shift from traditional design practices, expanding design's role into a multi-agent force.

Case Study 2: Strategic Foresight as an Agentic Design Competency

Episode: Human x AI Hybrid Intelligence

This episode examines the Strategic Foresight design competency embedded in the Service Design role shifting from a traditional design competency into an agentic capability within a collaborative innovation environment. Positioned as an intelligent agent, this competency engages with various roles, conducting trend investigations and identifying strategic opportunities to inform and drive solution development in dynamic contexts.

Context:

The episode takes place at the Innovation Center of Banco de Crédito del Perú (BCP), the largest bank in Peru and a subsidiary of Credicorp. Within this setting, a specific laboratory serves as the backdrop, where multidisciplinary teams are tasked with conducting strategic foresight to uncover emerging opportunities. In this process, the role of design is pivotal, acting as a bridge to identify needs, gaps, and opportunities, while fostering collaboration with roles such as developers and business leaders. This dynamic environment demands rapid

adaptation and seamless collaboration among team members to align foresight efforts with actionable innovation strategies.

Technical Aspects

- Participant: An AI developer with over 7 years of experience in coding, specializing in machine learning and the implementation of AI models.
- Tools and Software:
 - o Airtable Agents: AI-powered tools integrated with Airtable to automate tasks, streamline processes, and generate actionable insights by interacting directly with Airtable data.
 - o Azure Cloud Platform: The organization's cloud infrastructure, utilized to test and deploy Airtable Agents at scale. By hosting these agents in Azure, the organization ensures secure, scalable, and real-time testing of agent interactions and functionalities.
- AI Agent Training: The Service Design AI agent was trained on strategic foresight design competence, encompassing roles, goals, and tasks specific to this domain. This ensures the agent can effectively simulate and execute foresight processes, aligning with organizational innovation objectives.
- Cross-Role Training: In addition to its primary competency, the AI agent was trained to collaborate with and understand the responsibilities of other key roles, including Product Manager, Data Architect, AI Developer, and Software Architect. This cross-role training enables the agent to interact seamlessly within multidisciplinary teams, fostering collaboration and enhancing its adaptability in complex environments.
- Interaction workflow: The AI developer collaborates with the different AI agents through a co-creative process, iteratively refining content suggestions.

Objective of the Episode

- Analyze the process: Observe how the role of Service design and its strategic foresight competency in action evolves into agency through the coupling and decoupling process when interacting with different entities, humans and no humans.
- Identify intersections: Map the points where the design competency generates meaningful intersections with entities.

Activities

1. Introduction & onboarding: Explain the context and goal of the session.
2. Baseline task: Assign a task to the multi agent platform
3. Intersections map: Mapping how the strategic foresight competency interacts with various entities (other AI agents and humans) and observing how these interactions generate intersections that transform sub-competencies into agentic states.

Data collected

- Service Design Career Path
- Baseline task
- Intersection mapping
- Outputs and iterations
- Reflection log

Data samples:

Multi-Agent Front-End experience: The experiment involved simulating interactions between multiple AI agents and human participant

- Figure 12: This figure illustrates the baseline task, which involves defining a digital experience strategy tailored to current trends in Peru for small and medium-sized enterprises (SMEs). It provides an overview of the task objectives and its alignment with strategic foresight principles.
- Figure 13: This figure details the selection of specific intelligent agents tasked with collaborating with the human participant to accomplish the baseline task. It highlights the roles and capabilities of each agent, showcasing how they integrate .
- Figure 14: This figure captures the outputs generated during the interaction between intelligent agents and the human participant. It illustrates the iterative process, including the initial insights provided by the agents, the participant's feedback, and the refined solutions developed through subsequent iterations. This figure demonstrates the practical application of agentic collaboration in addressing the baseline task.

Dejale tareas al sistema multiagente

¿Qué título resume la tarea?

Define una estrategia de experiencia de

¿En que consiste la tarea?

Identifica tendencias, necesidades, oportunidades, propón iniciativas y priorízalas según impacto y esfuerzo

¿Qué agentes deben participar?

product manager arquitecto de software

diseñador de servicios ia developer

arquitecto de data

Haz seleccionado 0 agentes.

Ponlo a prueba

Figure 12: Baseline Task

Dejale tareas al sistema multiagente

¿Qué título resume la tarea?

Define una estrategia de experiencia de

¿En que consiste la tarea?

Identifica tendencias, necesidades, oportunidades, propón iniciativas y priorízalas según impacto y esfuerzo

¿Qué agentes deben participar?

product manager arquitecto de software

diseñador de servicios ia developer

arquitecto de data

Haz seleccionado 3 agentes.

Ejecutando agentes

Figure 13: Agents configuration

Ejecutando agentes		Nueva tarea
Agente	2024-11-14T04:32:37.000Z	Tipo: AgentFinish
de diseñador de servicios		
Input	Output	
Tarea:	Actualmente, mi presentación sobre la identificación de tendencias, necesidades, oportunidades, propuestas de iniciativas y su priorización según impacto y esfuerzo se encuentra en una etapa avanzada. He	
Payload:		
Agente	2024-11-14T04:32:37.000Z	Tipo: AgentAction
product manager		
Input	Output	
Tarea:	Actualmente, mi presentación sobre la identificación de tendencias, necesidades, oportunidades, propuestas	
Ask question to co-worker		

Figure 14: Output sample

Key takeaways

- **Dynamic configuration of design competencies in action:** The coupling and decoupling mechanism forms the foundation for the operationalization of Design Agency, allowing for the dynamic configuration of design competencies in action. The specific combinations of agents and roles significantly influence the value that the design competency delivers. For example, the interaction between the Service Design Agent and the Product Manager was observed to be highly outcome-driven, focusing on measurable financial impact. In contrast, the interaction between the same agent and the Data Architect proved more complex, requiring extended conversations between the agents and the human participant to align objectives and bridge gaps in understanding.
- **Tailoring coupling and decoupling for optimal Design Agency:** The coupling and decoupling process is a cornerstone of operationalizing Design Agency, enabling design competencies to adapt dynamically to various contexts. However, we have learnt that this process is not a one-size-fits-all solution; its success hinges on tailoring configurations to the specific roles, tasks, and objectives involved in each scenario.

- **Strategic foresight as an agentic capability:** This episode further underscores the evolution of Strategic Foresight as a design competency into an agentic capability. At these intersections, foresight becomes a dynamic force, integrating insights from diverse roles such as Product Managers and AI developers. The interaction complexity seen in some configurations reflects the transitional nature of these competencies, as they adapt and align across interdisciplinary boundaries.

Indicative Findings

This research uncovers the concept of Design Agency, highlighting its potential to address complex challenges and relevance. This new approach allows us to think beyond traditional design frameworks and operating models, exploring adaptable and scalable concepts applicable to dynamic and uncertain environments..

1. Design Agency through Coupling and Decoupling Competencies

The findings reveal that Design Agency emerges from the ability of design competencies to dynamically couple and decouple. This enables flexibility and responsiveness, transforming design into a modular system that can adapt and reconfigure as needed.

2. Design Agency as a Scalable System of Collective Intelligence

We learned that when combined with advanced forms of intelligence and supported by coupling and decoupling processes, Design Agency enables the integration of knowledge at scale.

3. Design agency for value creation at scale through Parenting Advantage

This research identifies Design Agency as a transformative mechanism for driving value creation at scale across corporate ecosystems. By enabling design to evolve into a scalable, distributed capability, corporations open new possibilities for empowering subsidiaries to develop their own internal Design Agencies, tailored to their unique challenges and opportunities. This decentralized approach introduces a new potential for design to enable Parenting Advantage, where the corporation moves beyond governance to actively enhance the value creation capacity of its subsidiaries. A pivotal aspect of this finding is the concept of a marketplace of corporate intelligence, where shared design frameworks, tools, and resources are dynamically exchanged and customized across business units enabling value creation..

The model

The indicative findings led to the development of a new working model for Design Agency, refining our understanding of what a Design Agency is and, more importantly, what it can do. This new model marks a transition from a broad theoretical framework to a more actionable model for implementing Design Agency in practice.

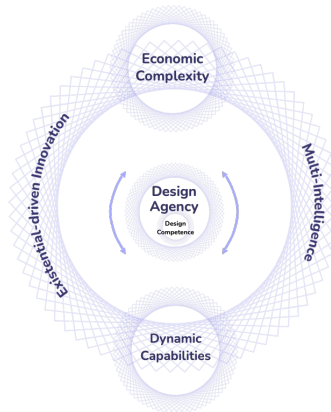


Figure 15: Theoretical model

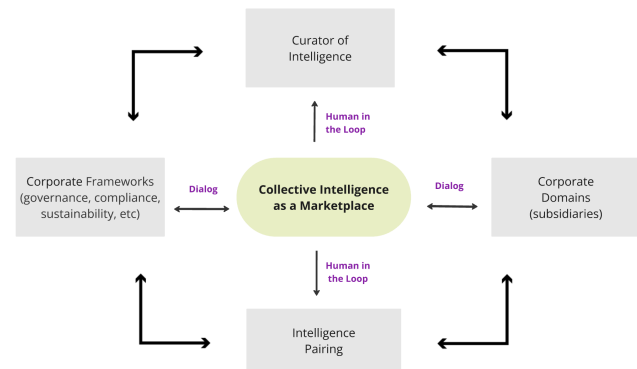


Figure 16: Design Agency Model

Early Theoretical Model (Left)

The theoretical model (Figure 15) was developed to explore the concept of Design Agency and to answer fundamental questions about its nature and practice.

- Key Components:
 - Design Agency at the Center: Positioned as the integration point for key theories: Dynamic Capabilities (D. Teece, 2017), Economic Complexity (R. Haussman 20090), and concepts such as Multi-Intelligence and Existential Threats. Together, these frameworks form a robust foundation for understanding how Design Agency operates within complex corporate ecosystems.
- Interrelated Forces:
 - Dynamic Capabilities: Emphasizes an organization's ability to adapt, orchestrate resources, and drive innovation in response to shifting environments. This provides the strategic agility needed to operationalize Design Agency.
 - Economic Complexity: Offers a lens to understand the interdependencies and knowledge-driven value creation inherent in

corporate ecosystems. This highlights the role of Design Agency as a knowledge broker.

- Multi-Intelligence Systems: Focuses on the interplay between human and artificial intelligence, showcasing how these hybrid systems collaborate to solve intricate challenges. Design Agency operates as a mediator, harnessing both intelligences to optimize outcomes.
- Existential-Driven Innovation: Recognizes that organizations often innovate in response to existential threats or challenges, such as technological disruption, sustainability demands, or market shifts.

These components collectively position Design Agency as a transformative force, capable of driving strategic alignment, fostering adaptability, and enabling innovation in increasingly complex and uncertain environment.

This broad model helped guide the initial research, including studies on coupling and decoupling, intersections, and episodes, each revealing the practical and theoretical boundaries of Design Agency.

New Working Model (Right)

This working model (Figure 16:) represents a significant evolution in the understanding of Design Agency, moving beyond a singular design competency to a multi-intelligence capability. It reframes Design Agency as a dynamic system embedded within a Marketplace of Collective Intelligence, where human and AI agents collaborate and optimize design configurations. This shift opens new opportunities for corporations in achieving value creation at scale.

Key Components:

- Collective Intelligence as a Marketplace at the core: The marketplace represents a dynamic system where humans and design competencies, enhanced by LLM capabilities, collaborate and adapt to meet the specific needs of conglomerate's subsidiaries. Borrowing Minsky's Society of Mind (1986), it functions as a distributed intelligence network, generating tailored solutions while driving a new vision of

Parenting Advantage for corporate strategy. This approach creates a competitive edge by aligning collective intelligence with strategic goals across the conglomerate.

- New functions: These new functions act as mechanisms of orchestration and governance rather than standalone roles. They enable the efficient and adaptive operation of the model, ensuring alignment with corporate objectives while addressing the specific needs of subsidiaries.
 - **Curator of Intelligence:** This function provides strategic oversight, ensuring the quality, relevance, and alignment of marketplace interactions with the organization's overarching goals. It acts as a governance mechanism to maintain focus and coherence within the system.
 - **Intelligence Pairing:** A mechanism designed to efficiently match design competencies—whether human-driven or AI-augmented—with the specific needs of subsidiaries or corporate domains. This pairing ensures optimized resource utilization and the delivery of precise, contextualized solutions.
 - **Corporate Domains (Subsidiaries):** Represent the unique demands and challenges of the corporation's subsidiaries, functioning as "users" of the marketplace. These domains leverage on-demand access to design competencies, ensuring relevant and scalable solutions reach all levels of the organization.
 - **Corporate Frameworks:** Provide the structural rules, policies, and standards that guide the marketplace. This function ensures alignment with governance, sustainability, and scalability objectives, maintaining ethical and strategic consistency across the system.
- The model emphasizes the importance of keeping humans in the loop at critical points of interaction, such as decision-making, governance, and iterative refinement. This ensures that ethical considerations, empathy, and strategic alignment remain central.

The working model positions Design Agency as a knowledge broker, integrating insights and competencies across diverse domains to address complex challenges. This integration bridges human and machine intelligence, fostering adaptability and innovation. Anchored in Teece's Dynamic Capabilities framework, Design Agency offers a practical model for integrating and orchestrating resources and knowledge, enabling organizations to navigate uncertainty and create strategic value in complex systems.

Conclusions

Design is undergoing a profound transformation, moving beyond the limitations of human cognition to embrace algorithmic capabilities and large language models (LLMs) in corporate environments. At the core of this transformation lies Design Agency, a multi-intelligent approach that elevates design from a human-centric, product-focused competency to a strategic, adaptive force capable of addressing the complexities of modern business contexts. While Design Thinking has successfully transitioned design into a team-oriented, competency-based approach, its reliance on human cognitive and manual skills limits its ability to tackle increasingly complex, data-intensive challenges. The shift from Design Thinking to Design Agency within a multi-intelligent framework redefines the role of design, allowing it to function not only as a human-powered competency but as a hybrid, agentic force that merges design competencies, human intelligence, and artificial intelligence. This evolution positions design as an enabler of innovation, adaptability, and strategic value creation in corporate ecosystems.

Next steps

As we progress with our research plan, the immediate priority is to finalize data collection and analysis, ensuring a thorough examination of our findings. These findings will be synthesized and structured into distinct, focused articles, aligning with the decision to pursue the multi-article track. Immediate priorities:

- Conducting Episodes for Case Study 1 and Case Study 2: Continue the execution of episodes for Case Study 1: Leveraging Agency-Driven UX Writing to Transform User Experience, and Case Study 2: Developing an Agentic Design Competency in Research for Innovation.
- For the Multi-Intelligence course, we are currently in the phase of gathering the materials developed throughout the weeks of the course, including prototypes created during the sessions. Participants are focusing on structuring the report, synthesizing the findings, and organizing insights derived from the research. Materials related to the course can be found in the associated folder.
- **Triangulation of Findings:** Integrate the findings from these episodes with the insights gained from previous research stages. This triangulation will ensure that the research is comprehensive and that the findings align with and enhance the latest iteration of the Design Agency Model.

- **Writing the Articles:** Begin drafting the articles based on the research findings and insights. Each article will focus on a distinct aspect of Design Agency, synthesizing the evidence gathered through the case studies and triangulation process.. The goal is to target a defense by the end of Fall 2025

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